

Extreme Values for Function of Two Variables

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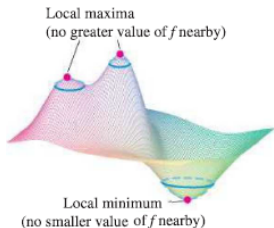
local maximum value of f if

$$f(a, b) \geq f(x, y)$$

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$$f(a, b) \leq f(x, y)$$

for all domain points (x, y) in an open disk centered at (a, b) .



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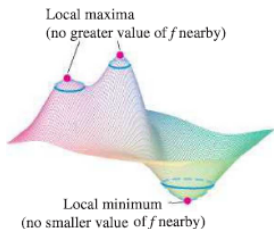
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- Local maxima correspond to mountain peaks
- Local minima correspond to valley bottoms.
- At such points the tangent planes, when they exist, are horizontal.
- Local extrema are also called **relative extrema**.

First Derivative Test for Local Extreme Values

Theorem

If $f(x, y)$ has a local extreme value at an interior point (a, b) of its domain and if the first partial derivatives exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Critical Points

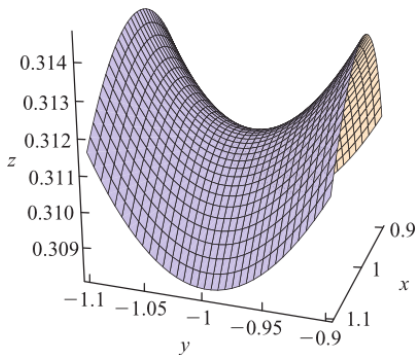
An interior point of the domain of a function $f(x, y)$ where both

- 1 f_x and f_y are zero or
- 2 where one or both of f_x and f_y do not exist

Saddle Point

The point $P(a, b, f(a, b))$ is a saddle point of $z = f(x, y)$ if (a, b) is a critical point of f and if every open disk centered at (a, b) contains points (x, y) in the domain of f for which

- $f(x, y) < f(a, b)$ and some points and $f(x, y) > f(a, b)$ for other points.



Checking for saddle points

$$f_{xx}f_{yy} - (f_{xy})^2 < 0$$

If $f(x, y) = x^2 - 4xy + y^3 + 4y$, find critical points of f .

- ① For Critical Points, set $f_x = f_y = 0$

$$\begin{aligned}f_x &= 2x - 4y = 0 \\f_y &= 3y^2 - 4x + 4 = 0\end{aligned}$$

we have $x = 2y$ from first equation,
using it in second $3y^2 - 8y + 4 = 0$
 $y = 2$ & $y = \frac{2}{3} \implies x = 4$ & $x = \frac{4}{3}$

- ② The Critical Points are $(4, 2)$ and $(\frac{4}{3}, \frac{2}{3})$
- ③ To find discriminant $D = f_{xx}f_{yy} - (f_{xy})^2$, calculate f_{xx} , f_{xy} and f_{yy}

$$f_{xx} = 2, \quad f_{xy} = -4 \quad \& \quad f_{yy} = 6y \implies D = 12y - 16$$

Critical Point	Value of D	Saddle Point
$(4, 2)$	$D(4, 2) = 8 > 0$	No
$(\frac{4}{3}, \frac{2}{3})$	$D(\frac{4}{3}, \frac{2}{3}) = -8 < 0$	Saddle point

Second Derivatives Test

Suppose that $f(x, y)$ has continuous second-order partial derivatives in some open disk containing the point (a, b) and that

$$f_x(a, b) = f_y(a, b) = 0.$$

Define the discriminant D for the point (a, b) by

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

- (i) If $D(a, b) > 0$ and
 - ❶ $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b)
 - ❷ $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b)
- (iii) If $D(a, b) < 0$, then f has a saddle point at (a, b) .
- (iv) If $D(a, b) = 0$, then no conclusion can be drawn.

If $f(x, y) = x^2 - 4xy + y^3 + 4y$, find local extrema and saddle point of f .

① For Critical Points, set $f_x = f_y = 0$

$$2x - 4y = 0, \quad 3y^2 - 4x + 4 = 0 \quad \xRightarrow{x=2y} \quad y = 2 \text{ \& } y = \frac{2}{3}$$

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Critical Point	Value of D	Value of f_{xx}	Conclusion
$(4, 2)$	$D(4, 2) = 8 > 0$	$f_{xx}(4, 2) = 2 > 0$	$f(4, 2) = 0$ is a local minimum

If $f(x, y) = x^2 - 4xy + y^3 + 4y$, find local extrema and saddle point of f .

① For Critical Points, set $f_x = f_y = 0$

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$(\frac{4}{3}, \frac{2}{3})$	$D(\frac{4}{3}, \frac{2}{3}) = -8 < 0$	irrelevant	$(\frac{4}{3}, \frac{2}{3}, f(\frac{4}{3}, \frac{2}{3}))$ is a saddle point

- Find the local extreme values of $f(x, y) = x^2 + y^2 - 4y + 9$.
- Find all critical points of $f(x, y) = xe^{x^2/2 - y^3/3 + y}$

Absolute Maxima and Minima on Closed Bounded Regions

We organize the search for the absolute extrema of a continuous function $f(x, y)$ on a closed and bounded region R into three steps.

- 1 List the interior points of R
where f may have local maxima and minima and evaluate f at these points. These are the **critical points** of f .
- 2 List the boundary points of R
- 3 Look through the lists for the maximum and minimum values of f .

Find the absolute maximum and minimum values of

$$f(x, y) = 2 + 2x + 2y - x^2 - y^2$$

on the triangular region in the first quadrant bounded by the lines $x = 0, y = 0, y = 9 - x$.

- Critical Point is $(1, 1)$

$$f_x = 2 - 2x = 0, \quad f_y = 2 - 2y = 0, \implies f(1, 1) = 4$$

Find the absolute maximum and minimum values of

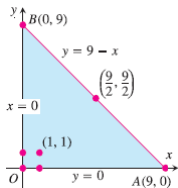
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- Boundary Points



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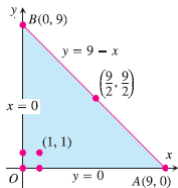
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$$f_x = 2 - 2x = 0, \quad f_y = 2 - 2y = 0, \implies f(1, 1) = 4$$

- Boundary Points



- 1 On $y = 0$: $f(x, y) = f(x, 0) = 2 + 2x - x^2$
- 2 On $x = 0$: $f(0, y) = 2 + 2y - y^2$
- 3 On $y = 9 - x$: $f(x) = -61 + 18x - 2x^2$

Finding Extreme Values of given function

$$f(x, y) = 2 + 2x + 2y - x^2 - y^2$$

We examine its behavior along $y = 0$, $x = 0$, and $y = 9 - x$.

Along $y = 0$

$$f(x, 0) = 2 + 2x - x^2$$

$$f'(x, 0) = 2 - 2x = 0$$

$$\Rightarrow \text{CP: } (1, 0)$$

Boundary: $(0, 0), (9, 0)$.

Along $x = 0$

$$f(0, y) = 2 + 2y - y^2$$

$$f'(0, y) = 2 - 2y = 0$$

$$\Rightarrow \text{CP: } y = 1 \text{ \& } x = 0$$

Boundary: $(0, 0), (0, 9)$.

Along $y = 9 - x$

$$f(x) = -61 + 18x - 2x^2,$$

$$f'(x) = 18 - 4x = 0 \Rightarrow x = \frac{9}{2}.$$

Boundary: $(0, 9), (9, 0)$

Critical: $(\frac{9}{2}, \frac{9}{2})$. $y = 9 - x$

Case	Point(s)	$f(x, y)$
$y = 0$	$(0, 0), (9, 0), (1, 0)$	2, -61 , 3
$x = 0$	$(0, 0), (0, 9), (0, 1)$	2, -61 , 3
$y = 9 - x$	$(0, 9), (9, 0), (4.5, 4.5)$	-61 , -61 , -20.5
At $(1, 1)$	—	4

Critical Points for $f(x, y)$

The point (a, b) is a critical point of the function $f(x, y)$ if (a, b) is in the domain of f and either

$$\frac{\partial f}{\partial x}(a, b) = \frac{\partial f}{\partial y}(a, b) = 0$$

or one or both of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ do not exist at (a, b) .

If $f(x, y)$ has a local extremum at (a, b) , then (a, b) must be a critical point of f .

Conditions for Extrema Using the Hessian

At a Critical Point (x_0, y_0) :

Local minimum

$$D > 0 \text{ and } \frac{\partial^2 f}{\partial x^2} > 0$$

Local maximum

$$D > 0 \text{ and } \frac{\partial^2 f}{\partial x^2} < 0$$

Saddle point $D < 0$

Inconclusive $D = 0$

Applications:

- Used in optimization problems.
- Crucial in machine learning, economics, and physics.

Locate and classify all critical points for $f(x, y) = 2x^2 - y^3 - 2xy$.

Step 1: Find Partial Derivatives

$$\frac{\partial f}{\partial x} = 4x - 2y, \quad \frac{\partial f}{\partial y} = -3y^2 - 2x$$

Step 2: Solve for Critical Points

$$\frac{\partial f}{\partial x} = 0 = \frac{\partial f}{\partial y}$$

$$4x - 2y = 0 \Rightarrow y = 2x,$$

$$-3y^2 - 2x = 0 \Rightarrow -12x^2 - 2x = 0.$$

$$x = 0, y = 0 \text{ and } x = -\frac{1}{6}, y = -\frac{1}{3}.$$

Critical points: $(0, 0)$ and $(-\frac{1}{6}, -\frac{1}{3})$.

Locate and classify all critical points for $f(x, y) = 2x^2 - y^3 - 2xy$.

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$$-3y^2 - 2x = 0 \Rightarrow -12x^2 - 2x = 0.$$

$$x = 0, y = 0 \text{ and } x = -\frac{1}{6}, y = -\frac{1}{3}.$$

Critical points: $(0, 0)$ and $(-\frac{1}{6}, -\frac{1}{3})$.

Step 3: Second Partial Derivatives

$$f_{xx} = 4, \quad f_{yy} = -6y, \quad f_{xy} = -2$$

Hessian determinant:

$$H = f_{xx}f_{yy} - (f_{xy})^2$$

At

$$(0, 0): H = (4)(-6(0)) - (-2)^2 = -4$$

$$H < 0 \Rightarrow \text{Saddle point}$$

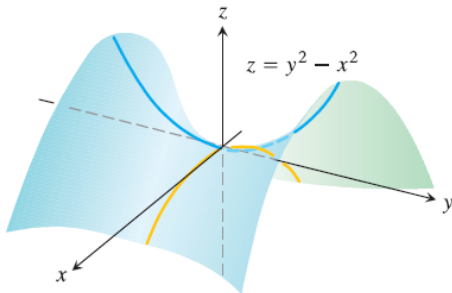
At $(-\frac{1}{6}, -\frac{1}{3}): H = 4$

$$f_{xx} > 0, \& H > 0 \Rightarrow \text{Local minimum.}$$

Final Classification of Critical Points

- $(0, 0)$: Saddle point.
- $(-\frac{1}{6}, -\frac{1}{3})$: Local minimum.

Find the local extreme values (if any) of $f(x, y) = y^2 - x^2$.



$$f_x = -2x, f_y = 2y \quad f_x = 0 = f_y \implies (x, y) = (0, 0)$$

$H = f_{xx}f_{yy} - (f_{xy})^2 < 0$ is a saddle point

$$f_{xx} = -2, f_{yy} = 2, f_{xy} = 0 \implies H = -4 < 0$$

Locate all relative extrema and saddle points of the function:

$$f(x, y) = 4xy - x^4 - y^4.$$

Step 1: Compute 1st-Order Partial Derivatives

$$f_x = 4y - 4x^3, \quad f_y = 4x - 4y^3.$$

$$x = (x^3)^3 = x^9 \implies x(x^8 - 1) = 0.$$

$$x = 0, x = \pm 1,$$

$$y = 0, y = \pm 1$$

Critical points:

- Set $f_x = 0$:
 $4y - 4x^3 = 0 \implies y = x^3.$
- Set $f_y = 0$:
 $4x - 4y^3 = 0 \implies x = y^3.$
- Solve $y = x^3$ and $x = y^3$:

Critical points:

$$(0, 0), \quad (1, 1), \quad (-1, -1).$$

Step 2: Compute Second-Order Partial Derivatives

$$f_{xx} = -12x^2, \quad f_{yy} = -12y^2, \quad f_{xy} = 4.$$

Hessian determinant:

$$H = f_{xx}f_{yy} - (f_{xy})^2 = (-12x^2)(-12y^2) - 4^2 = 144x^2y^2 - 16.$$

Step 3: Analyze Critical Points

Point	f_{xx}	f_{yy}	f_{xy}	H	Classification
(0, 0)	0	0	4	-16	Saddle point
(1, 1)	-12	-12	4	128	Relative maximum
(-1, -1)	-12	-12	4	128	Relative maximum

Relative maxima:

$$(1, 1) \quad \text{and} \quad (-1, -1).$$

Saddle point:

$$(0, 0).$$

Hessian Matrix

The Hessian matrix of a function is a square matrix of second-order partial derivatives. It provides information on the curvature of f .

For a function $f(x, y)$, the Hessian matrix H is:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

Hessian Matrix - Example

For $f(x, y) = x^3 + 3xy + y^2$:

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = 3$$

$$H = \begin{bmatrix} 6x & 3 \\ 3 & 2 \end{bmatrix}$$

- The Hessian matrix helps determine the behavior of f near critical points.